Project Progress & Final Technical Report

Course: Managing Corporate Entrepreneurship (B.Tech 7th Semester, 2026)
Institution: Indian Institute of Information Technology, Allahabad (IIITA)
Instructor / Supervisor: Dr. Netranand Pathak
Project Topic: Topic 21 — Digital Twin-Based Early Warning System for Cold Chain Disruption Detection Using Anomaly Detection
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Repository: https://github.com/Krishna200608/coldchain-ews-twin
Live Deployed Dashboard: https://coldchain-ews-twin.streamlit.app/

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1. Executive Summary

Cold-chain logistics for pharmaceuticals, biologics, and perishable foods require strict thermal adherence. Existing commercial solutions rely on **static threshold alarms** that fire only after a critical temperature boundary has been breached for a sustained period, by which time cumulative thermal damage is often irreversible.

This project designs, evaluates, and deploys a **Digital Twin-Integrated Early Warning System (EWS)** that embeds real-time anomaly detection algorithms inside a discrete-event simulation of the cold-chain distribution process. Using 97,606 physical IoT sensor readings as a structural proxy, an unsupervised Isolation Forest and a deep sequential LSTM-Autoencoder were developed, trained, and integrated into a three-stage SimPy digital twin simulation loop (`Source -> Transit -> Destination`).

The project is **100% technically complete**. All models, simulation pipelines, batch-to-live cross-checks, automated test suites, and interactive public dashboard layers have been implemented, verified, and pushed to GitHub.

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2. Research Gap, Objectives, and Core Research Questions

The Academic Gap

Cold-chain literature predominantly focuses on **post-hoc shelf-life / spoilage prediction** (calculating quality degradation after damage has occurred) or basic static threshold exceedance. Embedding proactive anomaly detection into an event-driven digital twin to provide quantifiable **early warning runway (Δt_{lead})** before irreversibility occurs remains underexplored.

Project Objectives

1. Develop an event-driven digital twin modeling the asynchronous streaming dynamics of cold-chain transit.
2. Embed classical unsupervised (Isolation Forest) and deep sequential (LSTM-Autoencoder) anomaly detectors into the digital twin loop.
3. Quantify and benchmark detection lead time against traditional static threshold alarms across diverse disruption categories.

Resolution of Research Questions (At a Glance)

Research Question	Empirical Findings	Status
RQ1: Early-Warning Lead Time *Can an anomaly model embedded in a digital twin flag disruptions before irreversibility, and how early?*	**Category-Specific Early Warning:** • **Drift Disruptions:** Isolation Forest achieved substantial early warnings of **+64.0 minutes** (`Out`) and **+11.0 minutes** (`In`) before irreversible failure. • **Step Disruptions:** All detectors alarmed concurrently (0.0 min lead time); ML provides zero lead-time advantage for sudden spikes. • **Flatlines:** The naive threshold baseline completely missed frozen signals; Isolation Forest detected both variance collapses. • **LSTM Lag:** The LSTM was late by **14.0 minutes** on `out_drift_009` due to gradual error adaptation.	**Resolved** (Table 1)
RQ2: Classical vs. Deep Learning *How do classical tree ensembles compare against deep sequential autoencoders on irregular telemetry?*	**Precision vs. Recall Trade-off:** • **Isolation Forest:** High sensitivity tripwire (Recall: 80.0% Out, 91.3% In; FPR: 45.6% Out, 27.0% In). • **LSTM-Autoencoder:** Highly conservative and precision-favoring on `In` series (Precision: 13.11%, FPR: 2.85%, AUC-PR: 0.3852), but near-chance on `Out` series (AUC-PR: 0.0049).	**Resolved** (Table 2)
RQ3: Digital Twin Simulation Fidelity *Does embedding models into a live event loop cause silent divergence from offline batch evaluation?*	**100% Exact Live/Batch Parity:** SimPy 3-stage simulation replayed 97,603 readings in 6.03 s CPU time (>1.9M$\times$ real time) with **0.00×10^0 s timing discrepancy**. Across 30 injections $\times$ 3 monitors, all 42 applicable test evaluations matched batch results bit-for-bit (0 mismatches), verified via buffer pre-warming (D33).	**Resolved** (`twin_crosscheck_report.csv`)

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3. Relevant Literature Review & Benchmark Foundations (Recent Papers 2023–2026)

Our research is grounded in recent literature spanning cold-chain digital twins, industrial IoT streaming, and machine-learning anomaly detection:

Study	Publication Venue	Core Contribution	Direct Relevance to Our Project
Vukšić et al. (2026)	*Applied Sciences* (MDPI)	Hybrid physics-based digital twin + convolutional autoencoder for reefer containers.	Reports median lead time of **~68 minutes**, providing independent corroboration for our **64-minute** early warning on drift disruptions.
Xie et al. (2025)	*PLOS ONE*	Improved Isolation Forest (subsampling + cross-grouping) for multi-sensor cold-chain fleet telemetry.	Benchmarks tree-based streaming anomaly detection in cold-chain logistics; contextualizes our single-channel point-wise F1 metrics.
Meng et al. (2024)	*Journal of Food Engineering* (Elsevier)	Cost-effective box-level digital twin temperature modeling + pallet alarms.	Exemplifies the static threshold-alarm paradigm that our proactive, model-embedded early warning system benchmarks against.
Wu et al. (2023)	*Journal of Industrial Information Integration* (Elsevier)	Internet-of-Everything and Digital-Twin-enabled service platform for cold-chain logistics.	Establishes architectural precedent for embedding AI models into live digital-twin data pipelines at platform scale.
Defraeye et al. (2023)	*Computers and Electronics in Agriculture / J. Food Eng.*	Comprehensive review of digital twins in food cold chains and postharvest operations.	Identifies the fundamental research gap: transitioning from post-hoc quality assessments to live, event-driven anomaly early-warning systems.

4. Data Strategy and Structural Proxy Notice

- **Dataset Source:** Kaggle `atulanandjha/temperature-readings-iot-devices` (`IOT-temp.csv`, 6.63 MB, 97,606 raw readings).
- **Timeframe:** July 28, 2018 to December 8, 2018 (133 continuous days, 27,920 unique minute timestamps).
- **Sampling Dynamics:** Highly irregular inter-arrival gaps (Mean: 117.82 s, Median: 0.0 s, Std: 5,446.3 s, Coefficient of Variation $\text{CoV} = 46.23$, Maximum outage gap: 11.57 days).
- **Structural Proxy Framing:** In keeping with strict scientific integrity, this dataset is recognized as a generic IoT office installation (`room_id/id = "Room Admin"`), not operational refrigerated fleet cargo. It is used exclusively as a **structural proxy** to evaluate streaming pipelines, irregular sampling, and integration architecture without making unverified food-safety claims (AD Log D5, D6, D10).

5. Methodology and System Architecture

1. **Ingestion & Causal Preprocessing (`src/preprocessing.py`):**
    - Corrected critical day-first timestamp parsing (`dayfirst=True`, preventing 48.8% silent parse failures).
    - Separated ambient (`Out`, 77,260 rows) and enclosure (`In`, 20,345 rows) streams.
    - Derived causal temporal features (arrival gap Δt_k, trailing rolling means and standard deviations, verified free of look-ahead leakage in D22).
2. **Controlled Synthetic Anomaly Protocol (`src/anomaly_injection.py`):**
    - Seeded 30 controlled perturbations (`RANDOM_SEED = 42`): 10 Step spikes, 10 Linear drifts, and 10 Frozen flatlines (15 Out / 15 In).
    - Applied a strict chronological 70% Train / 30% Test split (`cutoff_ts = 2018-10-29 11:10:48`).

    - 24 injections fall on the train split (used only for demonstrations); **all formal evaluation is restricted strictly to the 6 out-of-sample test injections** (D26).
3. **Detection Models:**
    - **Fixed Baseline (`src/baseline_and_labels.py`):** Static upper-bound threshold at train-period $\mu + 2.0\sigma$. Irreversibility timestamp (t_{irr}) defined as a sustained 10-minute exceedance.
    - **Isolation Forest (`src/isolation_forest_model.py`):** Classical unsupervised tree partition ensemble fit on train-only data with `contamination='auto'`.
    - **LSTM-Autoencoder (`src/lstm_prep.py`, `colab/lstm_train.ipynb`, `src/lstm_evaluation.py`):** Bivariate sequential reconstruction architecture (window $L=30$, step $S=5$), trained on Google Colab T4 GPU, with causal last-timestep MSE scoring against a train-only 2.0σ threshold.
4. **SimPy Discrete-Event Digital Twin (`src/digital_twin.py`, `src/twin_monitors.py`):**
    - Three-stage pipeline (`Source -> Transit -> Destination`) driving simulation progress using real inter-arrival gaps.
    - Streaming monitors (`BaselineMonitor`, `IFMonitor`, `LSTMMonitor`) embedded into the Destination sink.
    - Silent buffer pre-warming (D33) primes sliding buffers prior to visible replay windows, guaranteeing zero cold-start latency.

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6. Experimental Results and Performance Benchmark

Table 1: Early-Warning Lead Time Comparison (Test Split Injections)

$$\Delta t_{\text{lead}} = t_{\text{irreversibility}} - t_{\text{alarm}} \quad (\text{positive = early warning; negative = late})$$

Injection ID	Series	Fault Type	Baseline Alarm	IF Alarm	LSTM Alarm	Baseline Lead Time	IF Lead Time	LSTM Lead Time
`out_step_001`	Out	Step	`13:25`	`13:25`	`13:25`	0.0 min	0.0 min	0.0 min
`out_drift_009`	Out	Drift	`03:03`	`01:59`	`03:17`	0.0 min	**+64.0 min**	**-14.0 min (Late)**
`out_flatline_012`	Out	Flatline	None	`21:09`	`21:03`	Undefined	Alarm (No Irrev)	Alarm (No Irrev)
`out_flatline_013`	Out	Flatline	None	`01:22`	None	Undefined	Alarm (No Irrev)	Missed
`in_step_018`	In	Step	`17:14`	`17:14`	`17:14`	0.0 min	0.0 min	0.0 min
`in_drift_023`	In	Drift	`08:07`	`07:56`	`08:07`	0.0 min	**+11.0 min**	0.0 min

Table 2: Point-Wise Classification Metrics (Single Source of Truth, D35)

Guarded by automated regression assertion: `assert tp + fn == int(y_true.sum())`

Metric	IF — Out	LSTM — Out	IF — In	LSTM — In
Test Rows (Valid / Total)	13,372 / 13,372	13,343 / 13,372	3,767 / 3,767	3,738 / 3,767
Excluded (NaN Blind Spot)	0	29	0	29
True Anomaly Rows	65	65	23	23
True Positives (TP)	52	13	21	16
False Positives (FP)	6,067	2,722	1,010	106
True Negatives (TN)	7,240	10,556	2,734	3,609
False Negatives (FN)	13	52	2	7
Precision	**0.85%**	**0.48%**	**2.04%**	**13.11%**
Recall	**80.00%**	**20.00%**	**91.30%**	**69.57%**
F1 Score	**0.0168**	**0.0093**	**0.0398**	**0.2207**
False Positive Rate (FPR)	**45.59%**	**20.50%**	**26.98%**	**2.85%**
AUC-PR	**0.0676**	**0.0049**	**0.2371**	**0.3852**

7. Technical Deliverables Completed

Deliverable	Location in Repository	Description	Status
Pipeline Source Code	`src/*.py`	Modular scripts for ingestion, preprocessing, injections, ML training, evaluation, and SimPy simulation.	**Complete & Tracked**
Pre-Trained Weights	`models/lstm_{out,in}.h5`	Trained LSTM-Autoencoder weights exported from Colab GPU.	**Complete & Tracked**
Evaluation Artifacts	`data/processed/*.json`, `*.csv`	Serialized point-wise metrics, PR curves, and 3-way lead-time benchmarks.	**Complete & Tracked**
Interactive Dashboard	`dashboard/app.py`, `dashboard/components/`	Modular Streamlit web application with Guided Tour and Free Explore modes.	**Live on Web**
Academic Report	`docs/Report/Krishna_Sikheriya_ColdChain_Report.pdf`	12-chapter comprehensive LaTeX technical report formatted to IEEE / Springer standards.	**Complete (2.2 MB)**
Architectural Decision Log	`docs/AD_LOG.md`	35 formal entries documenting every architectural pivot, bug fix, and design decision.	**Complete (D1–D35)**
Dataset Submission	`Krishna_Sikheriya_IIT2023139_Topic21_Dataset.zip`	1.04 MB packaged archive containing `IOT-temp.csv` and detailed README.	**Ready for Portal**
Regression Test Suite	`tests/test_d33_prewarm.py`	Automated tests verifying pre-flight checks and buffer pre-warming (passing in 15.58 s).	**Passing**

8. Technical Readiness Assessment

Is the project complete technically?
Yes, 100% complete. There are zero unfinished code files, broken imports, missing weights, or failing tests. The entire pipeline reproduces from a clean state via standard terminal commands, and the interactive digital twin is deployed and publicly accessible online.

9. Presentation Readiness

For tomorrow's classroom presentation, the technical findings have been distilled into a crisp, 10-slide presentation deck documented in `docs/Presentation.md`, complete with slide structures, bullet points, and spoken cues.